

Trigonometry Mastery Blueprint

A Comprehensive Technical Reference Guide to Ratios, Core Identities, Quadrant Matrix, and Wave Mechanics

Trigonometry connects geometric triangle measurements to circular functions and wave mechanics. It forms the foundation for calculus, signal processing, spatial geometry, and computer graphics. This document contains key ratios, exact angle tables, identities, and analytical formulas.

1. FUNDAMENTAL RATIOS & RIGHT-ANGLE FRAMEWORKS

For a right-angled triangle featuring an acute baseline angle θ , with sides classified as **Opposite (O)**, **Adjacent (A)**, and **Hypotenuse (H)**:

Primary Ratios

- Sine: $\sin \theta = \text{Opp} / \text{Hyp} = O / H$
- Cosine: $\cos \theta = \text{Adj} / \text{Hyp} = A / H$
- Tangent: $\tan \theta = \text{Opp} / \text{Adj} = O / A = \sin \theta / \cos \theta$

Reciprocal Ratios

- Cosecant: $\csc \theta = 1 / \sin \theta = H / O$
- Secant: $\sec \theta = 1 / \cos \theta = H / A$
- Cotangent: $\cot \theta = 1 / \tan \theta = A / O$

2. EXACT VALUES MATRIX FOR STANDARD ANGLES

Exact value checkpoints derived analytically using special geometric boundaries (45° - 45° - 90° and 30° - 60° - 90° templates):

Ratio (θ)	0° (0)	30° ($\pi/6$)	45° ($\pi/4$)	60° ($\pi/3$)	90° ($\pi/2$)	180° (π)
$\sin \theta$	0	1/2	1/√2	√3/2	1	0
$\cos \theta$	1	√3/2	1/√2	1/2	0	-1
$\tan \theta$	0	1/√3	1	√3	Undefined	0
$\csc \theta$	Undefined	2	√2	2/√3	1	Undefined
$\sec \theta$	1	2/√3	√2	2	Undefined	-1
$\cot \theta$	Undefined	√3	1	1/√3	0	Undefined

3. THE CARTESIAN UNIT CIRCLE & QUADRANT SIGN AXIOMS

Mapping the unit circle ($x^2 + y^2 = 1$) sets coordinates as $(x, y) = (\cos \theta, \sin \theta)$. Signs adjust based on quadrants using the "All Silver Tea Cups" (ASTC) rule:

- Quadrant I (0° to 90°): All ratios are positive (+). | Reduction: θ
- Quadrant II (90° to 180°): Sine and Cosecant are positive (+); others are negative (-). | Reduction: $180^\circ - \theta$
- Quadrant III (180° to 270°): Tangent and Cotangent are positive (+); others are negative (-). | Reduction: $\theta - 180^\circ$
- Quadrant IV (270° to 360°): Cosine and Secant are positive (+); others are negative (-). | Reduction: $360^\circ - \theta$

4. FUNDAMENTAL PYTHAGOREAN & RECIPROCAL IDENTITIES

These essential equations are used to simplify expressions and solve trigonometric equations:

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

5. COMPOUND ANGLE COMPOUND FORMULATIONS

Formulas for the sum and difference of angles expand composite operations into modular individual outputs:

Sum Formulations • $\sin(A + B) = \sin A \cos B + \cos A \sin B$ • $\cos(A + B) = \cos A \cos B - \sin A \sin B$ • $\tan(A + B) = (\tan A + \tan B) / (1 - \tan A \tan B)$	Difference Formulations • $\sin(A - B) = \sin A \cos B - \cos A \sin B$ • $\cos(A - B) = \cos A \cos B + \sin A \sin B$ • $\tan(A - B) = (\tan A - \tan B) / (1 + \tan A \tan B)$
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6. DOUBLE & HALF-ANGLE EQUATION REDUCTIONS

These identities reduce powers and compress frequencies, which is essential for integration and wave analysis:

Double-Angle Matrices: • $\sin(2\theta) = 2 \sin \theta \cos \theta = (2 \tan \theta) / (1 + \tan^2 \theta)$ • $\cos(2\theta) = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta = (1 - \tan^2 \theta) / (1 + \tan^2 \theta)$ • $\tan(2\theta) = (2 \tan \theta) / (1 - \tan^2 \theta)$	
Half-Angle Formulations • $\sin(\theta/2) = \pm \sqrt{(1 - \cos \theta) / 2}$ • $\cos(\theta/2) = \pm \sqrt{(1 + \cos \theta) / 2}$	Power-Reduction Identities • $\sin^2 \theta = (1 - \cos 2\theta) / 2$ • $\cos^2 \theta = (1 + \cos 2\theta) / 2$

7. PRODUCT-TO-SUM & SUM-TO-PRODUCT CONVERSIONS

Product-to-Sum Formulations	Sum-to-Product Formulations
$2 \sin A \cos B = \sin(A + B) + \sin(A - B)$	$\sin x + \sin y = 2 \sin[(x+y)/2] \cos[(x-y)/2]$
$2 \cos A \sin B = \sin(A + B) - \sin(A - B)$	$\sin x - \sin y = 2 \cos[(x+y)/2] \sin[(x-y)/2]$
$2 \cos A \cos B = \cos(A + B) + \cos(A - B)$	$\cos x + \cos y = 2 \cos[(x+y)/2] \cos[(x-y)/2]$
$2 \sin A \sin B = \cos(A - B) - \cos(A + B)$	$\cos x - \cos y = -2 \sin[(x+y)/2] \sin[(x-y)/2]$

8. ADVANCED TRIANGLE PROPERTIES: LAWS OF SINES & COSINES

For any oblique non-right triangle with interior angles *A*, *B*, *C* mapped opposite to side constraints *a*, *b*, *c*:

• The Law of Sines: $a / \sin A = b / \sin B = c / \sin C = 2R$ (where <i>R</i> is the circumradius) • The Law of Cosines: $c^2 = a^2 + b^2 - 2ab \cos C \rightarrow \cos C = (a^2 + b^2 - c^2) / 2ab$ • Area of an Oblique Triangle: $Area = \frac{1}{2} ab \sin C = \frac{1}{2} bc \sin A = \frac{1}{2} ac \sin B$	
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